

BPE Tokenization of the Same Sentence Across Languages (Case 1: Byte Fallback)

(Fertility F = number of subword tokens produced per sentence)

English (Latin)

"The scientist discovered a new species"

The	scientist	discovered	a	new	species
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F = 6

German (Latin)

"Der Wissenschaftler entdeckte eine neue Art"

Der	Wissen	schaft	ler	ent	deck	te	eine	neue	Art
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F = 10

Amharic (Ge'ez)

"ሳይንቲስቱ አዲስ ዝርያ አገኘ" (LLaMA-3 Byte-Level BPE Fallback)

<0xE1>	<0x88>	<0xB3>	<0xE1>	<0x8B>	<0xAD>	<0xE1>	<0x8A>	<0x95>	<0xE1>	<0x89>	<0xB2>	...
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F = 38

Higher fertility (Fragmentation)

* Note: Amharic characters have no merges in LLaMA's BPE vocabulary; each 3-byte Ge'ez character decomposes into 3 raw UTF-8 byte tokens (<0x..>).